

TASK 2: EXPERIMENTATION AND UPLIFT TESTING

"In this project, I analyzed supermarket chip sales data to evaluate the effectiveness of a store trial. I first prepared and aggregated monthly sales data, then calculated key performance metrics such as total sales and customer count. Using correlation and magnitude distance, I identified Store 233 as the best control store for Trial Store 77. I compared the sales performance before and during the trial using visualizations and percentage differences. Finally, I performed a Welch Two Sample T-test to determine whether the observed improvement was statistically significant. The test returned a p-value of 0.2871, indicating that although some months showed higher sales, there was insufficient statistical evidence to confirm that the trial itself caused the improvement."

Project Objective

The objective of this project was to evaluate whether a trial conducted in selected stores led to an improvement in sales performance. To measure the impact of the trial, we compared the performance of a **Trial Store** with a similar **Control Store** using historical sales data.

Dataset

The dataset contains transactional data for chip sales, including:

- Store Number
- Transaction Date
- Customer ID

- Product Quantity
- Total Sales
- Product Details

The data covers monthly sales across multiple stores.

Step 1: Data Preparation

First, the dataset was imported into R using the **data.table** package.

The transaction date was converted into a monthly format (YearMonth) so that sales could be analyzed on a monthly basis.

Step 2: Monthly Performance Metrics

For each store and month, the following performance metrics were calculated:

- Total Sales
- Number of Customers
- Average Transactions per Customer
- Average Chips per Transaction
- Average Price per Unit

These metrics provide an overview of each store's monthly performance.

Step 3: Selecting Stores with Complete Data

Only stores having data for all twelve months were selected.

This ensured that every store had complete historical information before comparing their performance.

Step 4: Choosing the Control Store

The Trial Store was:

Store 77

To find the best Control Store, two similarity measures were calculated.

Correlation Score

Correlation measures how similarly two stores performed over time.

A higher correlation indicates that both stores follow a similar sales trend.

Magnitude Distance

Magnitude distance compares the actual sales values between stores.

Stores with smaller differences receive higher similarity scores.

Step 5: Final Control Store Selection

The correlation score and magnitude score were combined to calculate an overall similarity score.

The store with the highest final score was selected as the Control Store.

Result:

- Trial Store = **77**
- Control Store = **233**

Store 233 had the highest similarity with Store 77 before the trial period.

Step 6: Visual Comparison

Sales trends of the Trial Store, Control Store, and other stores were plotted using **ggplot2**.

The graphs showed that both Trial and Control stores followed similar trends before the trial, confirming that Store 233 was an appropriate control store.

Step 7: Trial Period Analysis

The trial period covered:

- February 2019
- March 2019
- April 2019
- May 2019
- June 2019

Sales of the Trial Store and Control Store were compared month by month.

The percentage difference between the two stores was calculated to observe the impact of the trial.

Step 8: Statistical Testing

A **Welch Two Sample T-test** was performed.

The purpose of the T-test was to determine whether the sales improvement observed during the trial period was statistically significant.

Result

- p-value = **0.2871**

Since the p-value is greater than **0.05**, the difference is **not statistically significant**.

This means there is not enough statistical evidence to conclude that the trial caused the increase in sales.

Final Conclusion

Store **233** was selected as the best Control Store because its historical performance closely matched that of Trial Store **77**.

Although Trial Store 77 recorded higher sales than the Control Store in some months—particularly **March and April 2019**—the statistical analysis showed that the improvement was **not statistically significant**.

The T-test produced a **p-value of 0.2871**, which is greater than the standard significance level of **0.05**.

Therefore, the available data does not provide sufficient evidence to conclude that the trial was responsible for the increase in sales.

Technologies Used

- **R Programming**
- **data.table**
- **ggplot2**
- **lubridate**
- **Statistical Analysis (Welch Two Sample T-test)**

```
setwd("C:/Users/myPC/OneDrive/Desktop/Quantium_Task2")
```

```
getwd()
```

```
list.files()
```

```
##### Libraries Load\
```

```
install.packages("data.table")
```

```
install.packages("ggplot2")
```

```
install.packages("tidyr")
```

```
library(data.table)
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
data <- fread("QVI_data.csv")
```

```
head(data)
```

```
library(lubridate)
```

```
data[, YEARMONTH := year(DATE) * 100 + month(DATE)]
```

```
install.packages("lubridate")
```

```
library(lubridate)
```

```
head(data)
```

```
library(data.table)
```

```
class(data)
```

```
library(lubridate)
```

```
data[, YEARMONTH := year(DATE) * 100 + month(DATE)]
```

```
head(data)
```

```
##Monthly Metrics (measureOverTime)
```

```
measureOverTime <- data[, .(
```

```
  totSales = sum(TOT_SALES),
```

```
  nCustomers = uniqueN(LYLTY_CARD_NBR),
```

```
  nTxnPerCust = uniqueN(TXN_ID) / uniqueN(LYLTY_CARD_NBR),
```

```
  nChipsPerTxn = sum(PROD_QTY) / uniqueN(TXN_ID),
```

```
  avgPricePerUnit = sum(TOT_SALES) / sum(PROD_QTY)
```

```
), by = .(STORE_NBR, YEARMONTH)][order(STORE_NBR, YEARMONTH)]
```

```
head(measureOverTime)
```

```
#Pre-Trial Data Filter
```

```
storesWithFullObs <- unique(
```

```
  measureOverTime[, .N, STORE_NBR][N == 12, STORE_NBR]
```

```
)
```

```
preTrialMeasures <- measureOverTime[
```

```
YEARMONTH < 201902 &
```

```
STORE_NBR %in% storesWithFullObs
```

```
]
```

```
head(preTrialMeasures)
```

```
length(storesWithFullObs)
```

```
##calculateCorrelation() Function
```

```
calculateCorrelation <- function(inputTable, metricCol, storeComparison) {
```

```
  calcCorrTable <- data.table(
```

```
    Store1 = numeric(),
```

```
    Store2 = numeric(),
```

```
    corr_measure = numeric()
```

```
  )
```

```
  storeNumbers <- unique(inputTable$STORE_NBR)
```

```
  for(i in storeNumbers){
```

```
    if(i != storeComparison){
```



```
corrValue <- cor(  
  
  inputTable[STORE_NBR == storeComparison, eval(metricCol)],  
  
  inputTable[STORE_NBR == i, eval(metricCol)]  
  
)
```

```
calcCorrTable <- rbind(  
  
  calcCorrTable,  
  
  data.table(  
  
    Store1 = storeComparison,  
  
    Store2 = i,  
  
    corr_measure = corrValue  
  
  )  
  
)
```

```
}
```

```
}
```

```
return(calcCorrTable)
```

```
}
```

```
trial_store <- 77
```

```
corr_nSales <- calculateCorrelation(  
  preTrialMeasures,  
  quote(totSales),  
  trial_store  
)
```

```
head(corr_nSales)
```

```
##### calculateMagnitudeDistance () Function
```

```
calculateMagnitudeDistance <- function(inputTable, metricCol, storeComparison){
```

```
  calcDistTable <- data.table(  
    Store1 = numeric(),  
    Store2 = numeric(),  
    YEARMONTH = numeric(),  
    measure = numeric()  
  )
```

```
storeNumbers <- unique(inputTable$STORE_NBR)
```

```

for(i in storeNumbers){

  if(i != storeComparison){

    calculatedMeasure <- data.table(

      Store1 = storeComparison,

      Store2 = i,

      YEARMONTH = inputTable[STORE_NBR == storeComparison, YEARMONTH],

      measure = abs(

        inputTable[STORE_NBR == storeComparison, eval(metricCol)] -

        inputTable[STORE_NBR == i, eval(metricCol)]

      )

    )

  }

  calcDistTable <- rbind(calcDistTable, calculatedMeasure)

}

}

minMaxDist <- calcDistTable[, .(

```

```
minDist = min(measure),
```

```
maxDist = max(measure)
```

```
), by = .(Store1, YEARMONTH)]
```

```
distTable <- merge(
```

```
  calcDistTable,
```

```
  minMaxDist,
```

```
  by = c("Store1", "YEARMONTH")
```

```
)
```

```
distTable[, magnitudeMeasure :=
```

```
  1 - (measure - minDist)/(maxDist - minDist)]
```

```
finalDistTable <- distTable[, .(
```

```
  mag_measure = mean(magnitudeMeasure)
```

```
), by = .(Store1, Store2)]
```

```
return(finalDistTable)
```

```
}
```

```
magnitude_nSales <- calculateMagnitudeDistance(
```

```
preTrialMeasures,  
  
quote(totSales),  
  
trial_store  
)  
  
head(magnitude_nSales)
```

```
corr_nCustomers <- calculateCorrelation(  
  
preTrialMeasures,  
  
quote(nCustomers),  
  
trial_store  
)  
  
head(corr_nCustomers)
```

```
magnitude_nCustomers <- calculateMagnitudeDistance(  
  
preTrialMeasures,  
  
quote(nCustomers),  
  
trial_store
```

)

head(magnitude_nCustomers)

corr_nCustomers <- calculateCorrelation(

preTrialMeasures,

quote(nCustomers),

trial_store

)

corr_weight <- 0.5

score_nSales <- merge(

corr_nSales,

magnitude_nSales,

by = c("Store1","Store2")

)

score_nSales[, scoreNSales :=

corr_weight * corr_measure +

(1 - corr_weight) * mag_measure]

```
head(score_nSales)
```

```
score_nCustomers <- merge(
```

```
  corr_nCustomers,
```

```
  magnitude_nCustomers,
```

```
  by = c("Store1", "Store2")
```

```
)
```

```
score_nCustomers[, scoreNCust :=
```

```
  corr_weight * corr_measure +
```

```
  (1 - corr_weight) * mag_measure]
```

```
head(score_nCustomers)
```

```
score_Control <- merge(
```

```
  score_nSales[, .(Store1, Store2, scoreNSales)],
```

```
  score_nCustomers[, .(Store1, Store2, scoreNCust)],
```

```
  by = c("Store1", "Store2")
```

```
)
```

```
score_Control[, finalControlScore :=
```

```
0.5 * scoreNSales +
```

```
0.5 * scoreNCust]
```

```
score_Control[order(-finalControlScore)][1:10]
```

```
control_store <- score_Control[order(-finalControlScore)][1, Store2]
```

```
control_store
```

```
##Step 15: Sales Trend Graph
```

```
library(ggplot2)
```

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
search()
```

```
ggplot()
```

```
ggplot(pastSales,
```

```
  aes(x = TransactionMonth,
```

```
      y = totSales,
```

```
      colour = Store_type)) +
```



```
geom_line(linewidth = 1)
```

```
ggplot(pastSales,  
       aes(TransactionMonth,  
            totSales,  
            colour = Store_type)) +  
geom_line(linewidth = 1) +  
labs(  
  title = "Total Sales by Month",  
  x = "Month",  
  y = "Total Sales"  
)
```

```
#####customer count graph
```

```
pastSales <- copy(measureOverTime)
```

```
pastSales[, Store_type := ifelse(  
  STORE_NBR == trial_store,
```

```

"Trial",

ifelse(STORE_NBR == control_store,

      "Control",

      "Other Stores")

)]

pastSales <- pastSales[

, .(totSales = mean(totSales)),

by = .(YEARMONTH, Store_type)

]

pastSales[, TransactionMonth := as.Date(

paste0(YEARMONTH %/% 100, "-", YEARMONTH %% 100, "-01")

)]

pastSales <- pastSales[YEARMONTH < 201903]

ggplot(pastSales,

      aes(TransactionMonth, totSales, colour = Store_type)) +

geom_line(linewidth = 1) +

labs(title = "Total Sales by Month")

```

```
###trialPeriod
```

```
trialPeriod <- measureOverTime[
```

```
  YEARMONTH >= 201902 &
```

```
  (STORE_NBR == trial_store | STORE_NBR == control_store)
```

```
]
```

```
trialPeriod
```

```
library(ggplot2)
```

```
trialPeriod[, Store_type := ifelse(STORE_NBR == trial_store,
```

```
  "Trial",
```

```
  "Control")]
```

```
trialPeriod[, Month := as.Date(
```

```
  paste(YEARMONTH %/% 100,
```

```
    YEARMONTH %% 100,
```

```
    "01",
```

```

      sep = "-")

    )]

ggplot(trialPeriod,

      aes(x = Month,

          y = totSales,

          colour = Store_type,

          group = Store_type)) +

  geom_line(linewidth = 1.2) +

  geom_point(size = 3) +

  labs(title = "Trial vs Control Store Sales",

        x = "Month",

        y = "Total Sales")

```

```
####trial effect (t-test) aur final conclusion
```

```
##percentage difference
```

```

trialSales <- measureOverTime[

  STORE_NBR %in% c(trial_store, control_store)

]

```

```

trialSales <- dcast(

  trialSales,

  YEARMONTH ~ STORE_NBR,

  value.var = "totSales"

)

names(trialSales) <- c("YEARMONTH", "Trial", "Control")

trialSales[, PercentageDiff := ((Trial - Control) / Control) * 100]

trialSales

##Trial months

trialSales[YEARMONTH >= 201902]

##Simple t-test

t.test(

  trialSales[YEARMONTH < 201902]$PercentageDiff,

  trialSales[YEARMONTH >= 201902]$PercentageDiff

)

```

```
library(ggplot2)

ggplot(trialPeriod,

       aes(x = TransactionMonth,

           y = totSales,

           colour = Store_type,

           group = Store_type)) +

  geom_line(linewidth = 1.2) +

  geom_point(size = 3) +

  labs(

    title = "Trial Period Sales Comparison",

    x = "Month",

    y = "Total Sales"

  ) +

  theme_minimal()
```

1. T-Test (Theory)

A **T-test** is a statistical method used to compare the means of two groups and determine whether the difference between them is statistically significant or occurred by chance.

In this project, the T-test was used to compare the percentage difference in sales between the **pre-trial period** and the **trial period** for the Trial Store and the Control Store.

The test helps answer the question:

Did the trial campaign actually improve sales, or was the observed difference simply due to random variation?

2. Statistical Significance

Statistical significance indicates whether the observed difference between two groups is likely to be real rather than caused by random chance.

The decision is based on the **p-value**.

- **p-value < 0.05:** The result is statistically significant, meaning the trial likely had a real impact.
- **p-value > 0.05:** The result is not statistically significant, meaning there is not enough evidence to conclude that the trial caused the observed difference.

In this analysis:

- **p-value = 0.2871**
- Since **0.2871 > 0.05**, the improvement in sales is **not statistically significant**.

This means the sales difference observed during the trial period could have occurred due to normal variation rather than the trial itself.

3. Final Conclusion

The analysis identified **Store 233** as the most suitable control store for **Trial Store 77** based on historical sales and customer behavior.

Although the Trial Store recorded higher sales than the Control Store in some months (especially **March and April 2019**), the statistical test produced a **p-value of 0.2871**, which is greater than the significance level of **0.05**.

Therefore, there is **insufficient statistical evidence** to conclude that the trial resulted in a significant increase in sales. The observed improvement may be due to normal business variation

rather than the trial campaign.

Overall Conclusion:

The trial showed positive sales performance in certain months, but the increase was **not statistically significant**, so the effectiveness of the trial cannot be confirmed with confidence based on the available data.